PWR_FLAG PWR_FLAG PWR_FLAG PWR_FLAG

GND VIN_BUCK +3V3 VDRV

PWR_FLAG: these nets are fed through passive parts (diodes, R4, L1, J1), so ERC needs a flag

POWER INPUT, FUSE, LOAD-DUMP TVS, REVERSE-POLARITY PROTECTION

J1 BATTERY 12V — VBAT_IN — GND

F1 15A — VBAT_IN — VBAT_F

D1 SMCJ20CA — VBAT_F — GND

Q1 IPD90P04P4L-04 — QG_REV — VBAT_F — VBAT_IN

R1 10k — QG_REV — V12 — GND

D2 BZT52C12 — V12 — QG_REV

C1 470uF 35V — V12 — GND

C2 470uF 35V — V12 — GND

C3 10uF 50V — V12 — GND

C4 10uF 50V — V12 — GND

MOTOR SWITCH: GATE DRIVER + LOW-SIDE N-MOSFET + FREEWHEEL DIODE

R4 220R — V12 — VDRV

D4 BZT52C12 — VDRV — GND

C5 1uF 25V — VDRV — GND

C6 100nF — VDRV — GND

U2 UCC27517A — PWM — VDRV — GATE_DRV — GND

R2 10R — GATE_DRV — GATE

R5 100k — PWM — GND

R3 10k — GATE — GND

Q2 IPD90N04S4L-04 — GATE — MOTOR_N — GND

D3 STPS20L45CG — V12 — MOTOR_N — MOTOR_N

J2 MOTOR — MOTOR_N — V12

3.3 V BUCK (12 V or USB)

D5 B360A — VIN_BUCK — V12

D6 B360A — VIN_BUCK — VBUS

C7 10uF 50V — VIN_BUCK — GND

C8 10uF 50V — VIN_BUCK — GND

C9 100nF 50V — VIN_BUCK — GND

R6 100k — VIN_BUCK — EN_BUCK

U1 LMR33630ADDA — VIN_BUCK — EN_BUCK — SW — BOOT — VCC_BUCK — FB — GND

C10 100nF — BOOT — SW

L1 6.8uH — SW — +3V3

R7 100k — +3V3 — FB

R8 43.2k — FB — GND

C12 22uF 10V — +3V3 — GND

C13 22uF 10V — +3V3 — GND

C11 1uF — VCC_BUCK — GND

ESP32-S3 MODULE, RESET/BOOT, BATTERY SENSE, LEDS, AUX

C14 10uF — +3V3 — GND

C15 100nF — +3V3 — GND

U3 ESP32-S3-WROOM-1 — +3V3 — EN — IO0 — PWM — IO5 — IO6 — USB_DM — USB_DP — VSENSE — LED — RXD0 — TXD0 — GND

R9 10k — +3V3 — EN

C16 1uF — EN — GND

SW1 RESET — EN — GND

R10 10k — +3V3 — IO0

SW2 BOOT — IO0 — GND

R13 100k — V12 — VSENSE

R14 18k — VSENSE — GND

C17 100nF — VSENSE — GND

R15 1k — LED — LED_A

D7 GREEN — LED_A — GND

J4 AUX — +3V3 — IO5 — IO6 — GND

R16 1k — +3V3 — PWR_A

D8 RED — PWR_A — GND

USB-C (PROGRAMMING / DEBUG)

J3 USB-C — VBUS — CC1 — CC2 — USB_DP — USB_DM — GND — SHIELD

U4 USBLC6-2SC6 — USB_DM — USB_DP — VBUS — GND

R11 5.1k — CC2 — GND

R12 5.1k — CC1 — GND

Motor: J2 pin 2 = +12 V (always on), pin 1 = MOTOR- (switched to GND by Q2).
PWM from ESP32 GPIO4, 20 kHz. R5 holds the gate driver input low while the ESP32 boots.
Battery sense: VSENSE = VBAT x 18k / 118k -> 14.4 V reads 2.20 V on GPIO1.
Q1 + D2: reverse-battery protection (P-MOSFET, gate clamped to 12 V).
D1 SMCJ20CA clamps load-dump spikes; everything on the 12 V side is rated >= 35 V.
Rated for motors up to 10 A continuous (15 A fuse). Use 2 oz copper.

H1 M3 H2 M3 H3 M3 H4 M3

Generated from design.py - board and schematic share one netlist
12 V car battery in (11-14.4 V), PWM low-side switched motor out, up to 10 A

Sheet: /		
File: motor_ctrl.kicad_sch		
Title: ESP32-S3 12 V Motor PWM Controller		
Size: A3	Date: 2026-10-26	**Rev: 1.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1